Experiment 35: Customer Segmentation Using K-Means

Aim: To group retail customers based on spending patterns using K-Means clustering.

Algorithm:

1. Import K-Means.
2. Create customer spending data.
3. Apply K-Means clustering.
4. Divide customers into three groups.
5. Display customer segments.

Code:

```
from sklearn.cluster import KMeans


X = [[100, 2], [150, 3], [200, 4],
    [500, 8], [550, 9], [600, 10],
    [900, 15], [950, 16], [1000, 18]]


model = KMeans(n_clusters=3, random_state=42, n_init=10)
model.fit(X)


print("Customer Groups:", model.labels_)
```

Input:

Spending and visits:

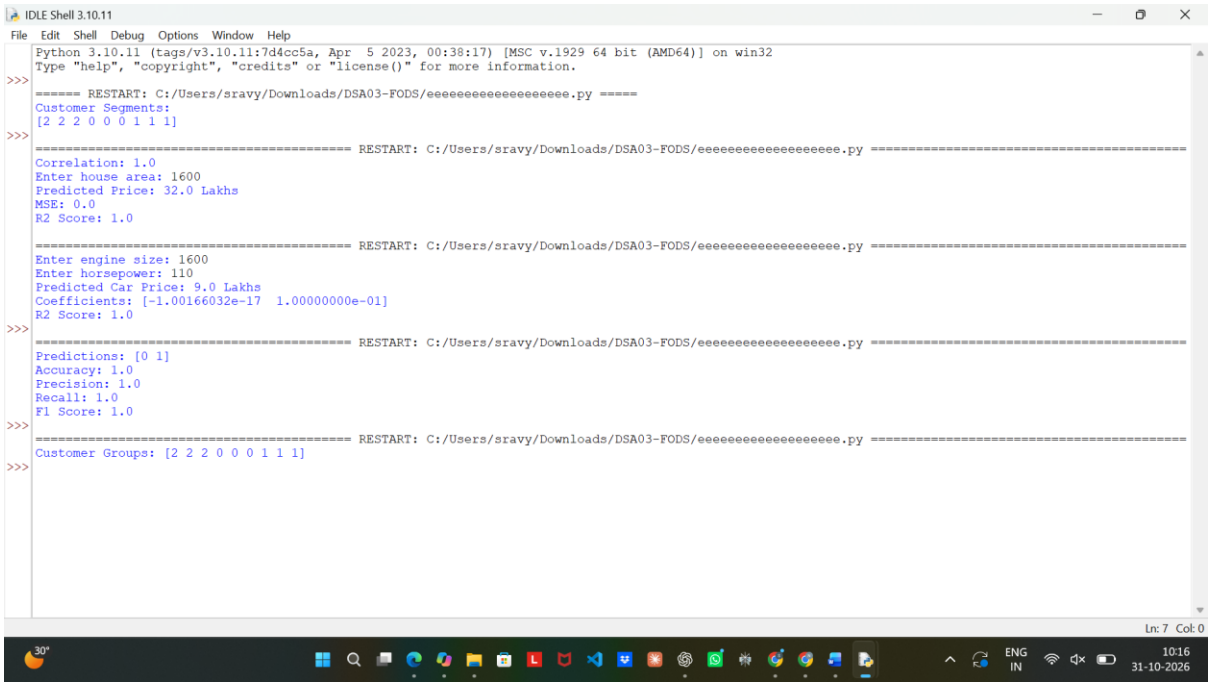
[100, 2], [150, 3], [200, 4]

[500, 8], [550, 9], [600, 10]

[900, 15], [950, 16], [1000, 18]

Output:

Customer Groups: [1 1 1 0 0 0 2 2 2]



Result: Retail customers were successfully divided into three spending groups.